

Assignment - 1

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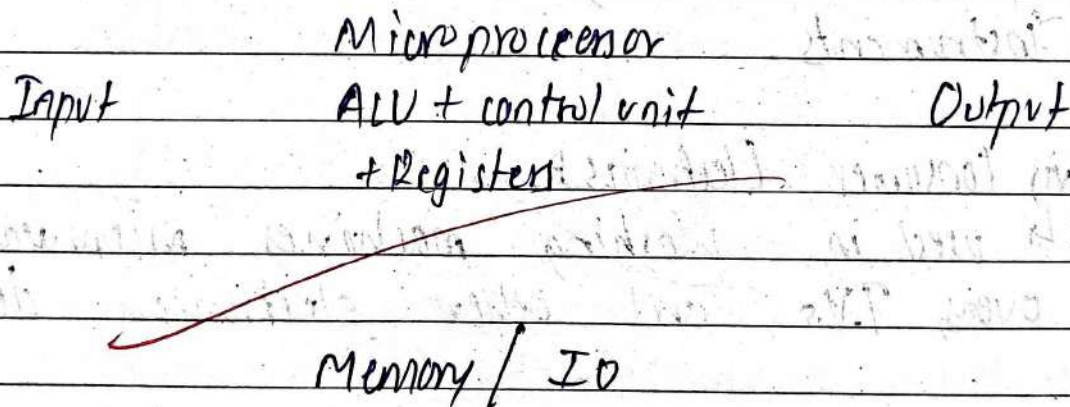
1) Define microprocessor. list out its different Applications

Ans A microprocessor is a programmable electronic device that performs arithmetic, logical and control operations.

- It is a CPU implemented on a single integrated circuit (IC) and executes instructions given by program.

ex. 8085, 8086, Pentium, ARM processor, etc

→ Basic diagram:



→ Applications:

i) personal computers:

↳ used in CPU of computers to perform calculation and execute programs.

ii) Industrial Control

↳ Used to control machines, robots and automated manufacturing systems

iii) Communication Systems

↳ Used in modems, routers, communication equipment, & telephone systems

iv) Automobiles

↳ Used in engine control, dashboard systems, safety systems and vehicle monitoring

v) Medical equipment

↳ Used in devices such as ECG machines, patient monitoring systems and medical instruments

vi) Consumer Electronics

↳ used in washing machines, microwaves, ovens, TVs and other electronic devices

vii) Office automation

↳ Used in printers, scanners, photocopies and other office equipment

viii) Embedded Systems

↳ Used as processing units in various dedicated electronic systems and control applications.

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 2) Explain different characteristics and features of microprocessor.

Ans i) Small size

↳ A microprocessor is fabricated on a single IC chip.

↳ ∴ it occupies very little physical space

ii) High speed

↳ It can execute many instructions in short period.

↳ processing speed mainly depends on its clock speed/frequency.

iii) Programmable

↳ Its operation can be changed by changing the program.

↳ The same processor can therefore be used for different applications.

iv) - low cost

↳ Integrated circuit technology reduces manufacturing cost.

↳ It makes microprocessor based system economical.

v) low power consumption:

↳ modern processors are designed to consume less power

vi) Versatility:

- ↳ It can perform arithmetic, logical, data transfer and control operations.
- ↳ It is used in many different type of electronic systems.

vii) Reliability:

- ↳ Many processing functions are into single IC chip.
- ↳ This reduces no. of hardware connections & improves reliability.

viii) Easy upgradation:

- ↳ system functionality can often be changed by changing program.
- ↳ Hardware can also be upgraded with a suitable processor.

3) Explain the History of Microprocessor

Ans The development of microprocessors took place through several generations.

i) First generation:

- ↳ microprocessors were introduced in early 1970s.
- ↳ They were mainly 4-bit processors such as Intel 4004.

ii) Second generation:

- ↳ 8-bit microprocessors were developed
- ↳ ex. 8080, 8085 & motorola 6800

iii) Third generation:

- ↳ 16-bit microprocessors were introduced
- ↳ examples include intel 8086 & 8088

iv) Fourth generation:

- ↳ 32-bit microprocessor became a common standard
- ↳ They provided high processing speed and large memory addressing capabilities.

v) Fifth generation:

- ↳ modern processor use 64-bit architecture, multiple cores, cache, memory and advanced technologies.
- ↳ they are used in computers, smartphones, servers and embedded systems.

h) Explain following terms:

i) Microcontrollers

↳ A microcontroller is an IC containing a CPU, memory and I/O ports on a single chip.

↳ ex. 8051, PIC, Arduino-based controllers

Microcontroller

- ↳ CPU
- ↳ RAM
- ↳ ROM
- ↳ memory
- ↳ I/O
- ↳ Timers

ii) Embedded Systems:

↳ An embedded system is a computer system designed to perform a specific function with a larger system.

↳ It generally contains a microcontroller or microprocessor along with software and hardware.

ex washing machine controller, traffic-light controller.

iii) Micro-computer

↳ A micro-computer is a small computer based on a processor as its CPU.

↳ It generally includes CPU, memory, I/O devices and storage.

ex. - personal computer, desktop computer.

5) Compare 8-bit & 16-bit microprocessor

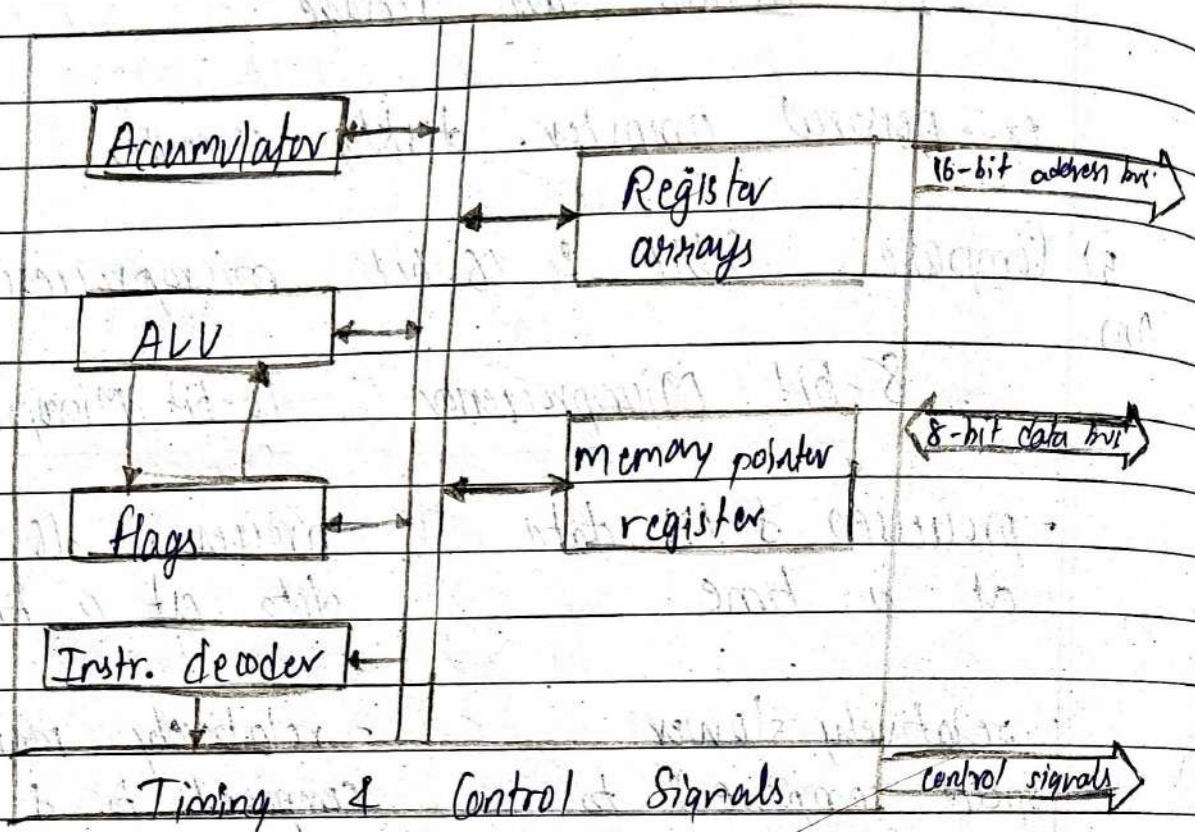
Aim

8-bit Microprocessor	16-bit Microprocessor
- processes 8-bit data at a time - relatively slower speed compared to 16-bit - usually smaller memory addressing - lower cost - simple control systems	- processes 16 bit data at a time - relatively faster speed compared to 8-bit - usually larger memory addressing - higher cost - more complex control system
ex. 8085 MPI	ex. 8086 MPI

6) Draw & explain Hardware Model and programming model of 8085 mp.

Ans 1) Hardware model of 8085 microprocessor-

- Shows internal hardware organization & functional units of 8085



a) Accumulator:-

↳ It is an 8-bit register which is used to perform all the arithmetic, logical, & many other ALU operations on.

↳ It stores result of all operations.

b) ALU:-

↳ ALU performs all the arithmetic & logical operations like addition, division, subtraction, AND, OR, NOT, XOR, etc.

c) Flag registers:

↳ These flags indicate the status of the result produced by ALU.

→ Flags:-

S - sign flag

Z - zero flag

AC - auxiliary carry

CF - carry flag

P - parity flag

d) Instruction Decoder

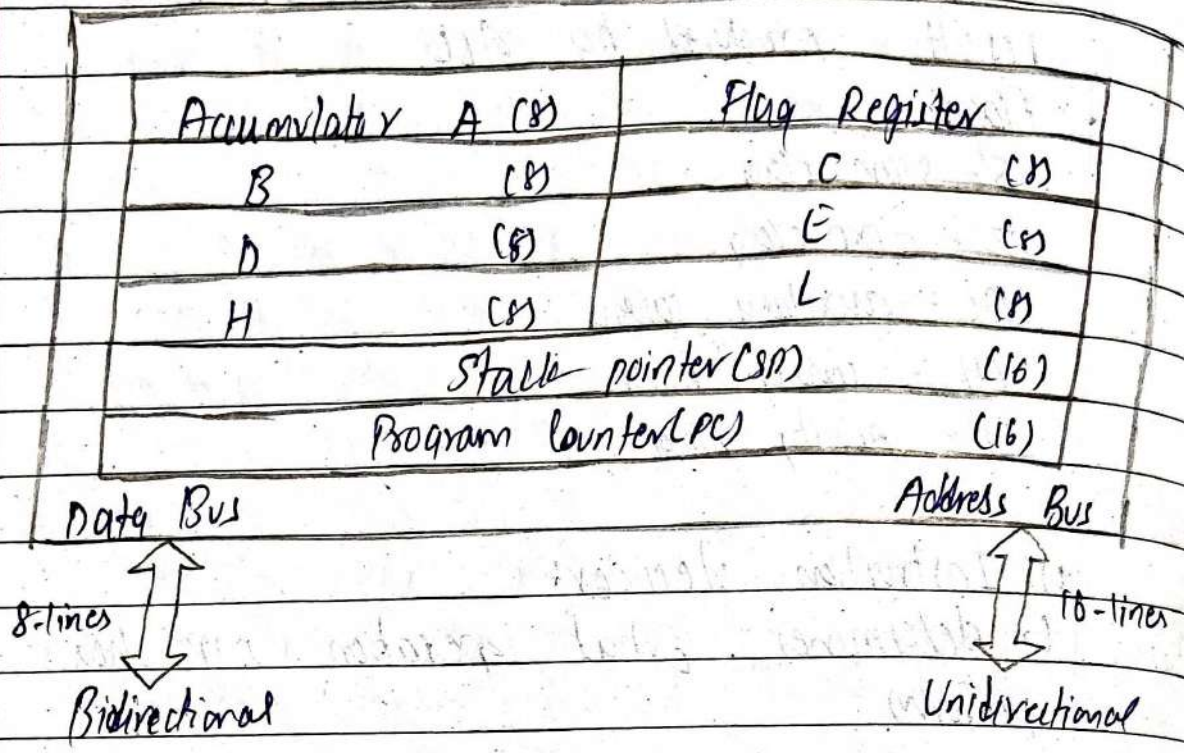
↳ determines what operation MP has to perform

e) Timing & control unit:

↳ Synchronises all the operations of the CPU.

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ii) Programming model of 8085:



- a) Accumulator (A) :-
- b) General purpose register
- c) Flag registers.
- d) Program Counter
- e) Stack pointer

Q7) Explain Bus system structure of 8085 MP

Ans The 8085 uses three types of buses for communication b/w the microprocessor memory and IO devices.

i) Address Bus

↳ The 8085 MP has a 16-bit address bus, allowing it to address up to 64 K_B of memory.

↳ It carries the address from the microprocessor to memory or IO devices

ii) Data Bus

↳ The 8085 has 8-bit bidirectional data bus

↳ It carries data b/w the microprocessor, memory & IO devices

iii) Control Bus

↳ It carries control & timing signals generated by the processor

↳ ex. RD, WR, ALE, IO/M

Q2 Explain different types of microprocessor operations.

Ans. A microprocessor performs different types of operations in its instruction set.

i) Data transfer operations:

↳ used to transfer data from one register to other

ex. MOV, MVI, LDA, STA

ii) Arithmetic operations

↳ used to perform addition, subtraction, increment, decrement

ex. ADD, SUB, INC, DEC

iii) Logical operations

↳ used to perform logical AND, OR, XOR, comparison, complement operations

ex. ANA, ORA, XRA, CMP, CMA

iv) Branching operations:

↳ used to change the normal sequence of program execution

ex. JMP, CALL, RET, conditional Jump

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v) Machine control operations

↳ used to control processor operation

ex. NOP, HLT, EI, DI

q) list and explain different instruction sets as per functionality and word size of 8085 μ P

Ans: Based on functionality:

a) Data transfer instructions:

→ transfer data b/w instructions registers memory & accumulators

ex. MOV, MVI, LDA, STR

b) Arithmetic instructions

↳ performs arithmetic operations

ex. ADD, INC, SUB, SBB

c) logical instructions

↳ perform logical and bit manipulation operations

ex. ANA, ORA, XRA, CMA

d) Branch Instructions

↳ change normal sequence of program execution

ex. JMP, CALL, RET, JC, JZ

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 v) Machine Control Instructions

↳ control the operation of the microprocessor
ex. NOP, HLT, EI, DI

iii) Based on word size:-

a) 1-byte instruction

↳ opcode itself

b) 1-byte instruction

↳ opcode itself contains all required information

ex. MOV A, B, ADD B

c) 2-byte instruction

↳ first byte contains opcode & second byte contains data

→ ex. MVI A, 32H

d) 3-byte instruction

↳ first byte contains opcode and other next two bytes contain a 16-bit address

ex. LDA 2050H

10) Define addressing mode. Explain various addressing modes in 8085 MP.

Ans Addressing mode specifies how an instruction identifies the operand on which the operation is performed.

→ 8085 supports five main addressing modes

i) Immediate Addressing

↳ Data is directly given in the instruction

ex. MVI A 25H

ii) Register Addressing

↳ operand is stored in a register

ex. MOV A B

iii) Direct Addressing

↳ The memory address of the operand is directly specified

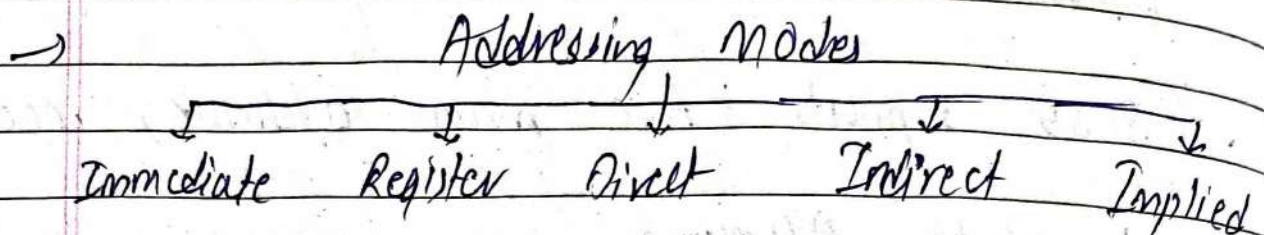
ex. LDA 2050H

iv) Register Indirect Addressing

↳ A register pair contains the address of operand

ex. MOV A M

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v) Implied / Implicit Addressing
is operand is implied by the
instruction itself.
ex. CMA



(ii) State Difference between:-

i) RLC and RAL

Ans

RLC

RAL

- rotate accumulator
left

- Rotate accumulator
through carry

- Bit D₇ moves to
Bit D₀ and carry

- Bit D₇ moves to
carry and old carry
moves to D₀

- carry is not used
as input

- carry is used
as input.

ex RLC

ex RAL

ii) RRC & RAR

Ans)

RRC

RAR

- Rotate accumulator right

- Rotate accumulator right through carry.

- Do moves to D₇ and carry

- Do moves to carry & old carry moves to D₇.

- carry is not used as input

- carry is used as input.

iii) PUSH and POP

Ans

PUSH

POP

- stores register - pair contents on the stack

- Retrieves data from the stack.

- stack pointer is decremented

- stack pointer is incremented.

- used to save temp. data

- used to retrieve previously stored data

ex PUSH B

ex POP R

12) Explain the following instructions with functionality, no. of bytes, machine cycle, Addressing mode and T-states.

Ans i) CALL

↳ calls a subroutine and stores the return address on the stack

→ Bytes: 3

→ Machine cycle: 5

→ T-states: 18

→ Addressing: Direct

ii) CPI

↳ compares immediate data with the accumulator

- Bytes: 2

- Machine cycles: 2

- T-states: 7

- Addressing: Immediate

iii) SHLD

↳ stores content of HL into two consecutive memory locations

- Bytes: 3

- Machine cycles: 5

- T-states: 16

- Addressing: Direct

iv) RAL

↳ Rotates accumulator left through the carry flag

- Bytes: 1
- Machine states: 1
- T-states: 4
- Addressing: Implied

v) CMA

↳ Complements all bits of the accumulator

- Bytes: 1
- Machine cycle: 1
- T-states: 4
- Addressing: Implied

vi) LDA

↳ loads the accumulator with data from the specified memory address

- Bytes: 1
- Machine cycles: 1/2
- T-states: 4/7
- Addressing mode: Register/Indirect

vii) M

↳ represent memory location whose address is contained in HL register pair

↳ Bytes / Machine cycle / T-states depend on use of M

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viii) LXI SP

↳ loads 16-bit immediate value into stack pointer

- Bytes: 3
- Machine cycles: 3
- T-states: 10
- Addressing modes: Immediate

ix) SHLD

↳ store contents of HL register into two consecutive memory locations

- Bytes: 3
- Machine cycles: 5
- T-states: 16
- Addressing modes: Direct

x) XCHG

↳ exchange HL pairs with DL pairs

- Bytes: 1
- Machine cycles: 1
- T-states: 4
- Addressing: Implied

13) Explain various flags in 8085 registers.
Ans The 8085 has a 5-bit flag register

- These flags indicate the result of arithmetic & logical operations

→ flag register:

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
S	Z	-	AC	-	P	-	OV

i) sign flag (S)

↳ It indicates the sign of the result.

- $S = 1$ when $MSB_{result} = 1$.

ii) zero flag (Z)

↳ It indicates whether the result is zero

↳ $Z = 1$ when the result of an operation is zero

iii) Auxiliary carry flag (AC)

↳ It is set when carry occurs from bit D₃ to D₄

↳ It is mainly used in BCD arithmetic operation

iv) Parity Flag (P)

↳ It indicates the parity of the result.

↳ $P=1$ when result contains even 1's.

v) Carry Flag (CH)

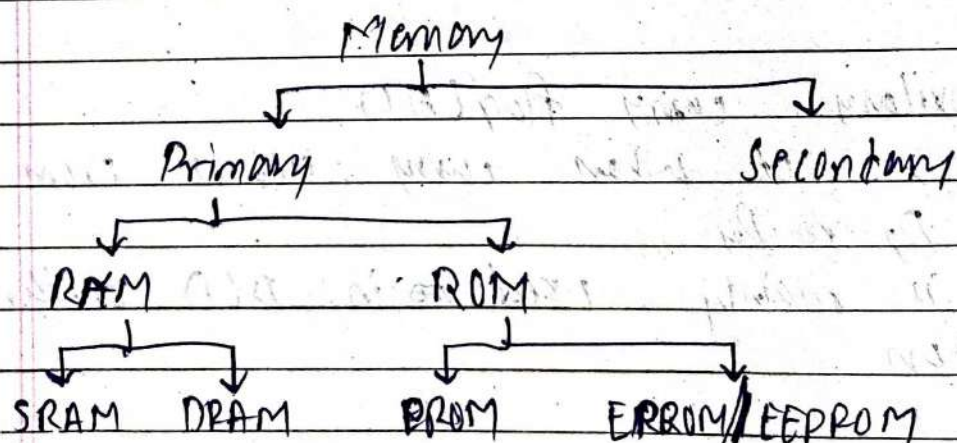
↳ It is set when an arithmetic operation generates a carry out.

↳ $CH=1$, When carry occurs, otherwise $CH=0$.

1m) What is memory? Explain different kinds of memory.

Ans Memory is a storage unit used to store programs, data, and results required to by the microprocessor.

→ It is broadly classified into primary memory & secondary memory.



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→ RAM: Random access memory

↳ used to read/write for temporary storage of data & program

↳ It is volatile memory, i.e. the contents are lost once the power is switched off

ex SRAM, DRAM

→ ROM: Read only memory

↳ ROM stores information that normally remains available even when power is removed

↳ It is non-volatile memory.

→ Types of ROM:

i) PROM: can be programmed once, after that it cannot be changed.

ii) EPROM: can be erased using UV light & programmed again

iii) EEPROM can be electrically erased & programmed.

iv) flash memory: type of electrically erasable non-volatile memory

- used in memory card, USB, embedded systems.

→ Secondary memory:

- used in long term storage of large data

- ex. HDD, CD, SSD, pen drive

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